Software Maintenance and Computer Aided Software Engineering (CASE)

(Lecture 12)

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Introduction

- Software maintenance:
  - any modifications to a software product after it has been delivered to the customer.
- Software maintenance is an important activity for many organizations.

Introduction

- Maintenance is inevitable for almost any kind of product.
- Most products need maintenance:
  - due to wear and tear caused by use.
- Software products do not need maintenance on this count.

Introduction

- Many people think
  - only bad software products need maintenance.
- The opposite is true:
  - bad products are thrown away,
  - good products are maintained and used for a long time.

Introduction

- Software products need maintenance for three reasons:
  - corrective
  - adaptive
  - perfective

Corrective

- Corrective maintenance of a software product:
  - to correct bugs observed while the system is in use.
  - to enhance performance of the product.

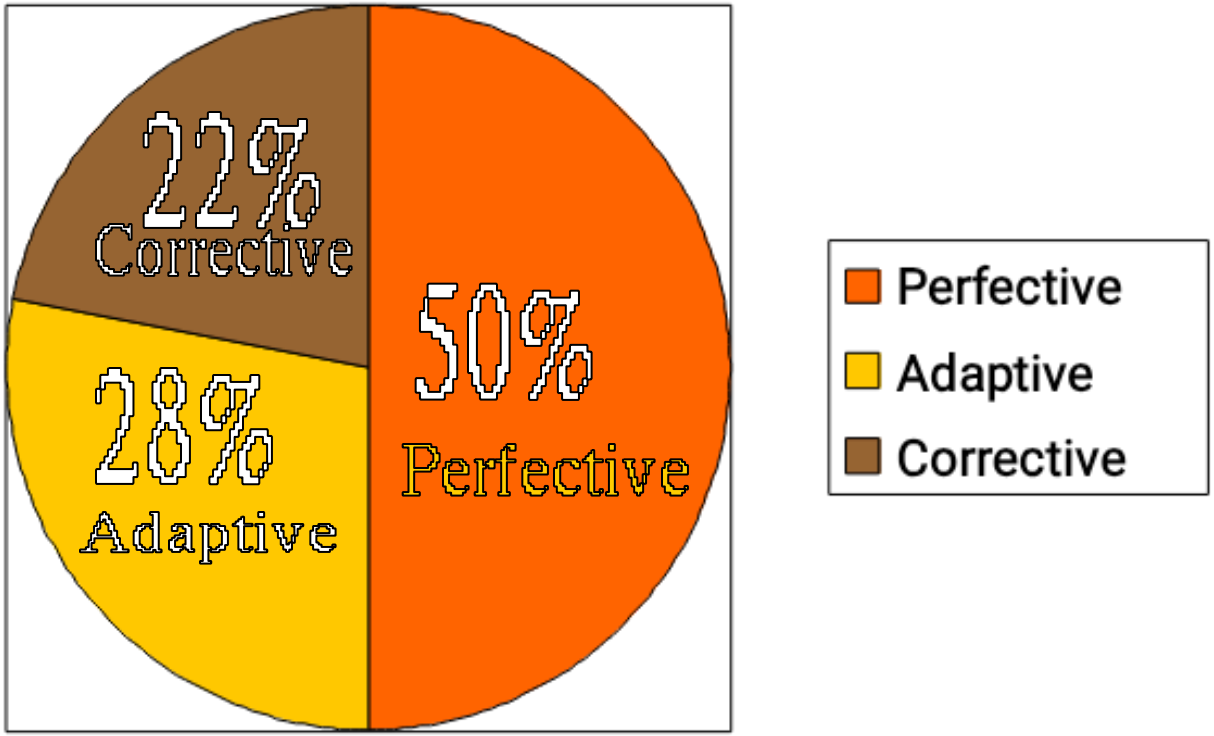
Adaptive

- A software product needs maintenance (porting) when customers:
  - need the product to run on new platforms,
    - or, on new operating systems,
  - need the product to interface with new hardware or software.

Perfective

- Perfective maintenance:
  - to support new features required by users.
  - to change some functionality of the system due to customer demands.

Maintenance Effort Distribution



Causes for maintenance

- During development:
  - Software not anticipated to last very long (e.g., Y2K problem).
- Rate of hardware obsolescence:
  - immortality of software products
  - maintenance necessary for software performing low-level functions.

Causes for maintenance

- Users want existing software to run on new platforms:
  - to run in new environments,
  - and/or with enhanced features.

Causes for maintenance

- Whenever other software it works with change:
  - maintenance is needed to cope up with the newer interface.
  - For instance, a software product may need maintenance when the operating system changes.

Software evolution

- Every software product continues to evolve after its development:
  - through maintenance efforts.
- Larger software products stay in operation for longer time:
  - because of high replacement cost.

Laws of Maintenance

- There will always be a lot of old software needing maintenance.
- Good products are maintained, bad products are thrown away.

Laws of Maintenance

- Lehman's first Law:
- "Software products must change continuously, or become progressively less useful."

Laws of Maintenance

- Lehman's Second Law
- "When software is maintained, its structure degrades,
  - unless active efforts are made to avoid this phenomenon."

Laws of Maintenance

- Lehman's Third Law:
- "Over a program's life time,
  - its rate of development is approximately constant."

Other Laws of Maintenance

- All large programs will undergo significant changes during operation phase of their life cycle,
  - regardless of *apriori* intentions.

Legacy code--- Major maintenance problems

- Unstructured code (bad programs)
- Maintenance programmers have:
  - insufficient knowledge of the system or the application domain.
  - Software maintenance has a bad image.
- Documentation absent, out of date, or insufficient.

Insufficient knowledge

- Maintenance team is usually different from development team.
  - even after reading all documents
    - it is very difficult to understand why a thing was done in a certain way.
  - Also there is a limit to the rate at which a person can study documents
    - and extract relevant information

Bad image of maintenance?

- Maintainers are skilled not only in writing code:
  - proficient in understanding others' code
  - detecting problems, modifying the design, code, and documentation
  - working with end-users

Maintenance Nightmares

- Use of gotos
- Lengthy procedures
- Poor and inconsistent naming
- Poor module structure
- Weak cohesion and high coupling
- Deeply nested conditional statements
- Functions having side effects

How to do better maintenance?

- Program understanding
- Reverse engineering
- Design recovery
- Reengineering
- Maintenance process models

Maintenance activities

- Two types of activities:
  - Productive activities:
    - modification of analysis, design, coding, etc.
  - Non-productive activities:
    - understanding system design, code, etc.

Software Reverse Engineering

- By analyzing a program code, recover from it:
  - the design and the requirements specification.

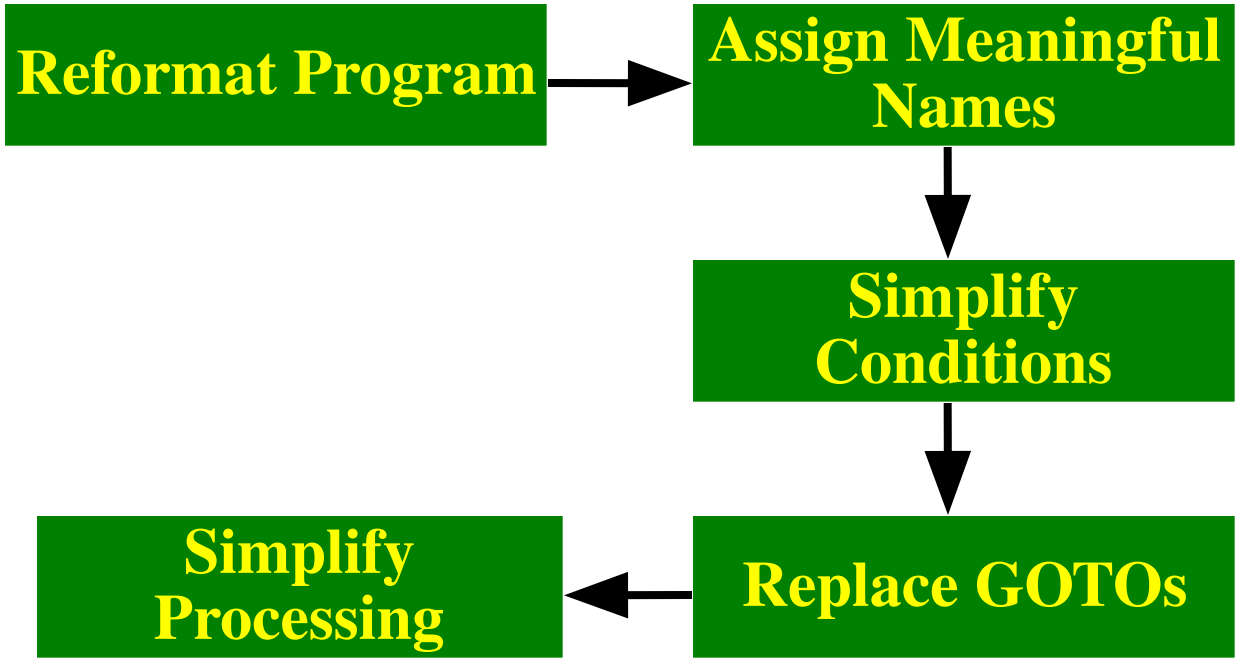
Software Reverse Engineering

- Reverse engineering is an important maintenance technique:
  - several existing software products are unstructured,
  - lack proper documentation,
  - were not developed using software engineering principles.

Software Reverse Engineering

- First carry out cosmetic changes to the code to improve:
  - readability,
  - structure,
  - understandability.

Cosmetic changes



Cosmetic Changes

- Reformat the program:
  - use any pretty printer program
  - layout the program neatly.
- Give more meaningful names to:
  - Variables, data structures, and functions.

Cosmetic Changes

- Replace complex and nested conditional expressions:
  - simpler conditional statements
  - whenever appropriate use case statements.

Software Reverse Engineering

- In order to extract the design:
  - fully understand the code.
- Automatic tools can be used to help derive:
  - data flow and control flow diagrams from the code.

Software Reverse Engineering

- Extract structure chart:
  - module invocation sequence and data interchange among modules.
- Extract requirements specification:
  - after thoroughly understanding the code.
  - design has been extracted.

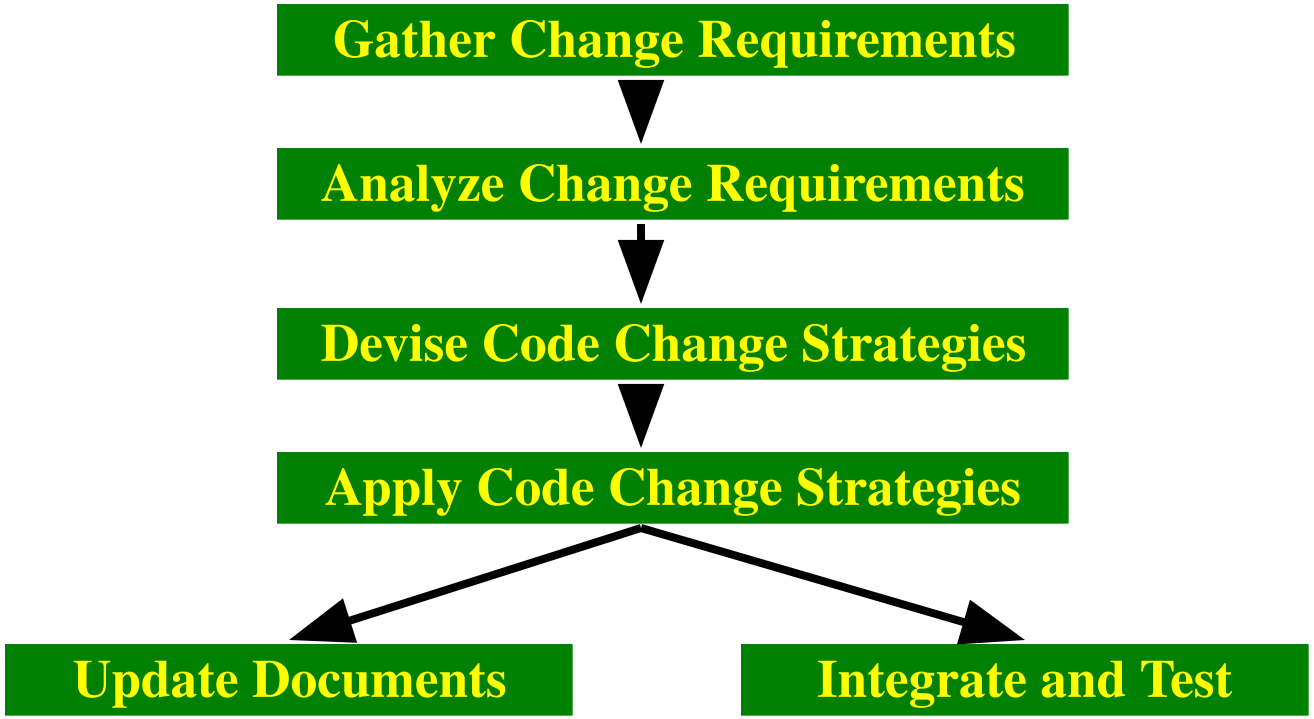
Software Maintenance Process Models

- Maintenance activities are not unique:
  - depend on the extent of modifications required,
  - also, depend on condition of the product:
    - how structured it is,
    - how well documented it is, etc.

Software Maintenance Process Model - 1

- When the required changes are small and simple:
  - the code can be directly modified
  - changes reflected in all relevant documents.
  - more elaborate activities are required when required changes are not trivial.

Software Maintenance Process Model - 1



Software Maintenance Process Model 1

- Start by gathering change requirements.
- Analyze change requirements
  - formulate strategies for code change.

Software Maintenance Process Model 1

- Formulating strategies for code change:
  - presence of few members of the original development team
    - helps in reducing cycle time,
    - especially for unstructured and inadequately documented code.

Software Maintenance Process Model 1

- Availability of a working old system at the maintenance site:
  - greatly helps the maintenance team
  - provides a good insight into the working of the old system
  - can compare the working of the modified system with the old system.

Software Maintenance Process Model 1

- Debugging the system under maintenance becomes easier:
  - program traces of both the systems can be compared to localize bugs.

Software Maintenance Process Model -2

- For complex maintenance projects, software reengineering needed:
  - a reverse engineering cycle followed by a forward engineering cycle.
  - with as much reuse as possible from existing code and other documents.

Maintenance Process Model 2

- Preferable when:
  - amount of rework is significant
  - software has poor structure.
- Can be represented by a reverse engineering cycle:
  - followed by a forward engineering cycle.

Software reengineering

- Many aging software products belong to this category.
- During the reverse engineering,
  - the old code is analyzed (abstracted) to extract the module specifications.

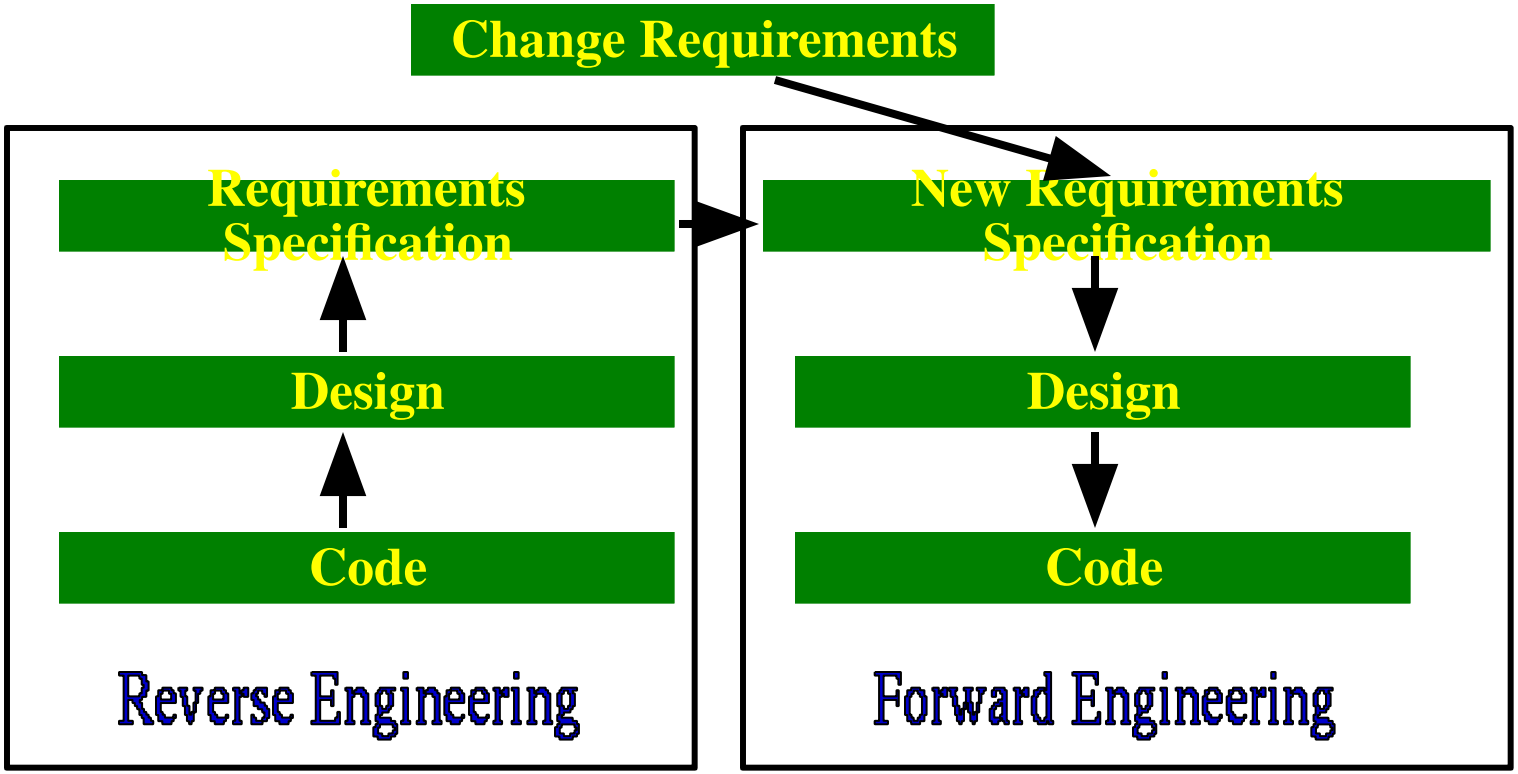
Software reengineering

- The module specifications are analyzed
  - to produce the design.
- The design is analyzed (abstracted)
  - to produce the original requirements specification.
- The change requests are then applied to the requirements specification:
  - arrive at the new requirements specification.

Software reengineering

- Forward engineering is carried out to produce the new code.
- During design, module specification, and coding:
  - substantial reuse is made from the reverse engineered products.

Process model for Software reengineering



Software reengineering

- Advantages of reengineering:
  - produces better design than the original product,
  - produces required documents,
  - often results in higher efficiency.

Software reengineering

- Efficiency improvements are brought about by better design.
  - However, this approach is more costly than the first approach.
- An empirical study indicates:
  - process 1 is preferable when amount of rework is no more than 15%.

Software reengineering

- Reengineering is preferable when:
  - amount of rework is high,
  - product exhibits high failure rate.
  - product difficult to understand.

Computer Aided Software Engineering (CASE)

- CASE tools help in software development and maintenance.
- CASE is a much talked about topic in software industries.

CASE and Its Scope

- CASE tool is a generic term:
  - denotes any form of automated support for software engineering.
- In a more restrictive sense:
  - a CASE tool <u>automates some software development activity</u>.

CASE and Its Scope

- Some CASE tools assist in phase-related tasks:
  - specification, structured analysis, design, coding, testing, etc.
- Other tools help non-phase activities:
  - project management and configuration management.

Objectives of CASE

- To increase productivity
- To help produce better quality software at lower cost.

CASE Environment

- Although individual CASE tools are useful:
  - true power of a tool set can be realized only when:
    - all CASE tools are integrated together.

CASE Environment

- Tools covering different stages of life cycle share information (data):
  - they should integrate through some central repository (store)
  - consistent view of development information.

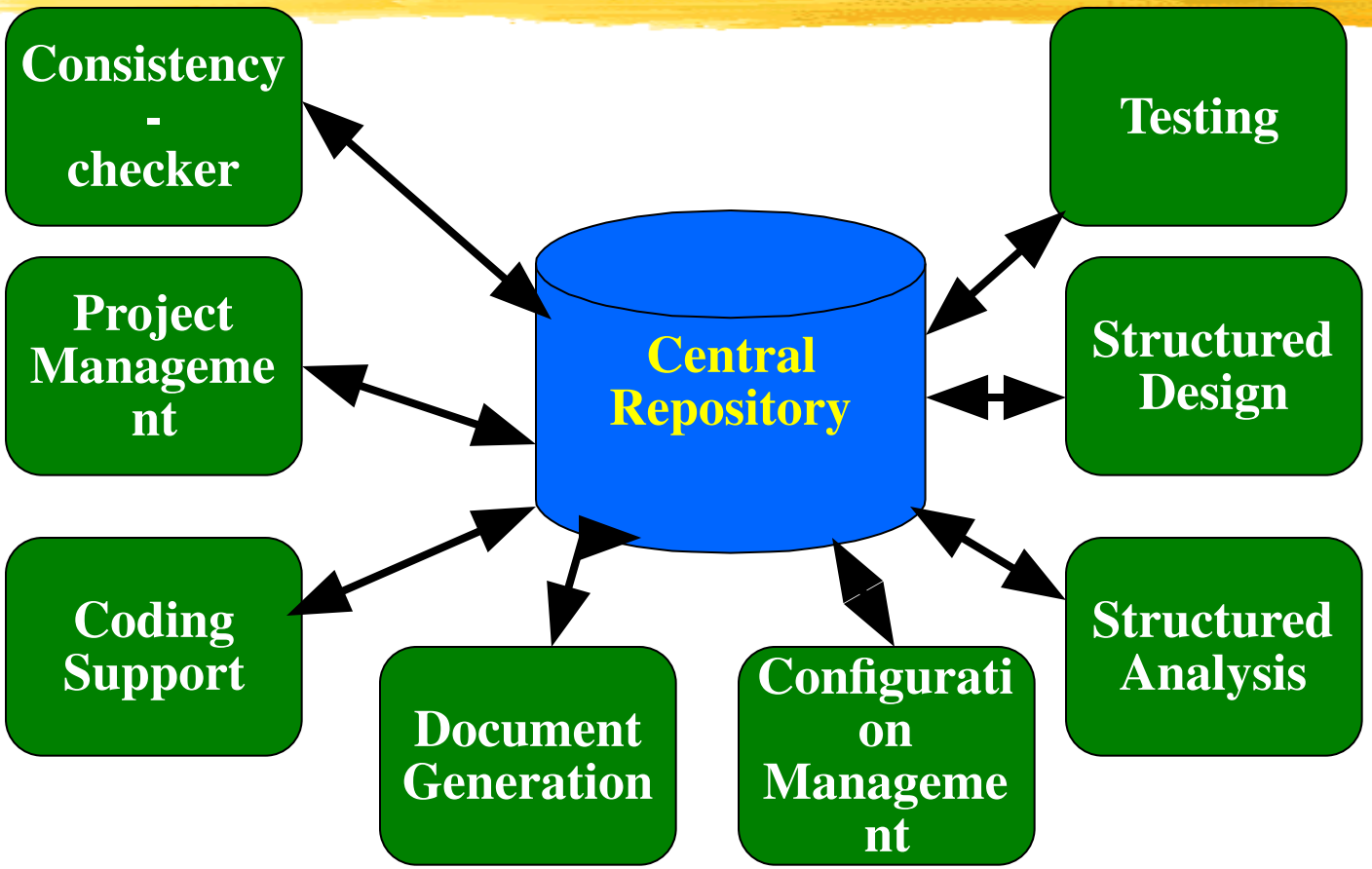
CASE Environment

- The central repository is the data dictionary:
  - contains definition of all composite and elementary data items.
  - through this repository all CASE tools share information.

Programming Environment

- A CASE environment helps:
  - automate step-by-step methodologies.
- In contrast to CASE environment:
  - a programming environment denotes tools supporting coding phase alone.

Schematic representation of architecture of CASE environment



Benefits of CASE

- A key benefit of using CASE environment:
  - cost saving through all developmental phases.
- Studies carried out to measure the impact of CASE usage:
  - cost saving between 30% to 40%.

Benefits of CASE

- Use of CASE tools leads to improvements in quality:
  - becomes easy to iterate through different software development phases.
  - chances of human error is reduced.
  - CASE tools help produce higher quality and consistent documents.

Benefits of CASE

- Data relating to a software product are maintained in a central repository:
  - redundancy in the stored data is reduced.
  - chances of inconsistent documentation is reduced.

Benefits of CASE

- CASE tools take drudgery out from software engineers' work:
  - engineers need not manually check balancing of the DFDs
  - easily draw diagrams and produce documentation, etc.

Benefits of CASE

- CASE tools lead to cost saving in software maintenance effort:
  - traceability and consistency checks,
  - systematic information capture during various development phases.

Benefits of CASE

- Introduction of a CASE environment:
  - impacts the style of working of engineers.
  - makes them oriented towards structured and orderly approach.

Prototyping Support

- Prototyping CASE tool:
  - often used in graphical user interface (GUI) development,
  - supports creating a GUI using a graphics editor.

Prototyping Support

- The user should be allowed to define:
  - data entry forms, menus and controls.
- It should integrate with the data dictionary of a CASE environment.

Structured Analysis and Design

- A CASE tool should:
  - support some standard structured analysis and design technique.
  - support easy creation of analysis and design diagrams.
  - should provide easy navigation through different levels of design and analysis diagrams.

Structured Analysis and Design

- The tool must support completeness and consistency checking.
- The tool should disallow inconsistent operations:
  - but, it is difficult to implement such a feature.

Code Generation

- As far as code generation is concerned:
  - expectations from a CASE tool is low.
- The CASE tool should support:
  - generation of module skeletons in one or more popular languages.
  - Another reasonable requirement is traceability from source code to design.

Code Generation

- It should automatically generate header information:
  - copyright messages,
  - brief description of the module,
  - author name and date of creation, etc.

Code Generation

- The tool should generate data records or structures automatically:
  - using data dictionary definitions.
  - It should generate database tables for relational database management systems.

Code Generation

- The tool should generate code for user interface from the prototype:
  - for X window and MS window based applications.

Testing Support

- Static and dynamic program analysis of programs.
- It should generate test reports in ASCII format:
  - which can be directly imported into the test plan document.

Desirable Features

- The tool should work satisfactorily
  - when many users work simultaneously.
- The tool should support windowing interface:
  - Enable the users to see more than one diagram at a time.
  - Facilitate navigation and switching from one part to the other.

Documentation Support

- The deliverable documents:
  - should be able to incorporate text and diagrams from the central repository.
  - help in producing up-to-date documentation.

Desirable Features

- The CASE tool should integrate
  - with commercially available desk-top publishing packages.
- It should be possible to export text, graphics, tables, data dictionary reports:
  - to DTP packages in standard formats such as PostScript.

Project Management

- It should support collecting, storing, and analyzing information on the software project's progress:
  - such as the estimated task duration,
  - scheduled and actual task start, completion date, dates and results of the reviews, etc.

External Interface

- The tool should allow exchange of information for reusability of design.
  - The information exported by the tool should preferably be in ASCII format.
- The data dictionary should provide
  - a programming interface to access information.

Reverse Engineering Support

- The tool should support:
  - generating structure chart, DFD, and data dictionary from source code.
  - should populate the data dictionary from source code.

Data Dictionary Interface

- Data dictionary interface should provide
  - viewing and updating the data definitions.
  - print facility to obtain hard copy of the viewed screens.
  - analysis reports like cross-referencing, impact analysis, etc.
  - it should support a query language.

Tutorial and Help

- Successful use of CASE tools:
  - depends on the users' capability to effectively use all supported features.
- For the first time users:
  - a computer animated tutorial is very important.

Tutorial and Help

- The tutorial should not be limited to teaching the user interface part only:
  - The tutorial should logically classify and cover all techniques and facilities.
  - The tutorial should be supported by proper documentation and animation.

Towards Next Generation CASE Tool

- An important feature of next generation CASE tools:
  - be able to support any methodology.
- Necessity of a CASE administrator for every organization:
  - who would tailor the CASE environment to a particular methodology.

Intelligent Diagramming Support

- Future CASE tools would
  - aesthetically and automatically lay out the diagrams.

Towards Next Generation CASE Tool

- The user should be allowed to:
  - integrate many different tools into one environment.
  - It is highly unlikely that any one vendor will be able to deliver a total solution.

Towards Next Generation CASE Tool

- A preferred tool would support tune up:
  - user would act as a system integrator.
  - This is possible only if some data dictionary standard emerges.

Customization Support

- The user should be allowed to define new types of objects and connections.
- This facility may be used to build some special methodologies.
- Ideally it should be possible to specify the rules of a methodology to a rule engine:
  - for carrying out the necessary consistency checks.

Summary

- We discussed some fundamental concepts in software maintenance.
- Maintenance is the mostly expensive phase in software life cycle:
  - during development emphasize on maintainability to reduce the maintenance costs.

Summary

- We discussed software reverse engineering:
  - extract design from code.
- Reengineering is a reverse engineering cycle:
  - followed by a forward engineering cycle

Summary

- Maintenance process models:
  - Process model for small changes
  - Process model for reengineering
- We also discussed:
  - applicability of process models to maintenance projects.

Summary

- We discussed important features of present day CASE tools:
  - and the emerging trends.
- Use of CASE tools is indispensable for large software projects:
  - where a team of software engineers work together.

Summary

- The trend is now towards:
  - distributed workstation-based CASE tools.
- We discussed some desirable features of CASE tools.